

Am2948/Am2949

Octal Three-State Bidirectional Bus Transceivers

DISTINCTIVE CHARACTERISTICS

- 8-bit bidirectional data flow reduces system package count
- 3-state inputs/outputs for interfacing with bus-oriented systems; PNP inputs reduce input loading
- $V_{CC} = 1.15V$ V_{OH} interfaces with TTL, MOS, and CMOS
- 48mA, 300pF bus drive capability; Low power – 8mA per bidirectional bit
- Am2948 has inverting transceivers; Am2949 has non-inverting transceivers — both have separate **TRANSMIT** and **RECEIVE** Enables
- Bus port stays in hi-impedance state during power up/down

GENERAL DESCRIPTION

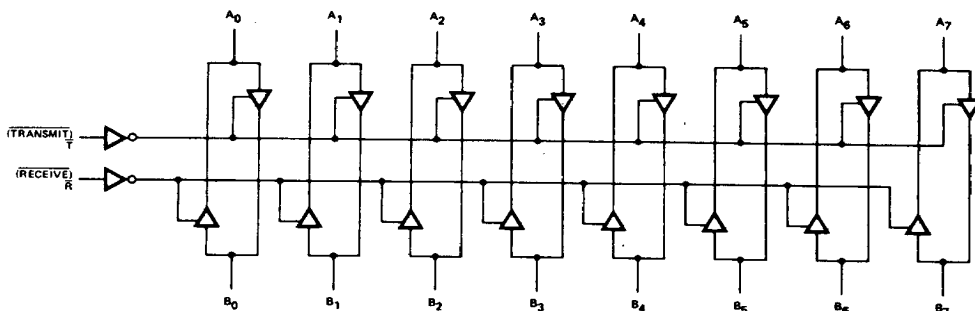
The Am2948 and Am2949 are 8-bit, 3-state Schottky transceivers. They provide bidirectional drive for bus-oriented microprocessor and digital communications systems. Straight through bidirectional transceivers are featured, with 24mA drive capability on the A ports and 48mA bus drive capability on the B ports. PNP inputs are incorporated to reduce input loading.

Separate **TRANSMIT** and **RECEIVE** Enables are provided for microprocessor system with separated read and write control bus lines.

The output high voltage (V_{OH}) is specified at $V_{CC} = 1.15V$ minimum to allow interfacing with MOS, CMOS, TTL, ROM, RAM, or microprocessors.

BLOCK DIAGRAM

Am2949



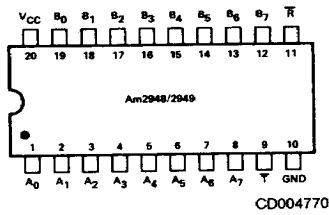
BD002500

Am2948 has inverting transceivers.

05407A

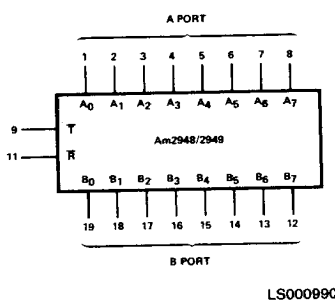
CONNECTION DIAGRAM Top View

D-20-1

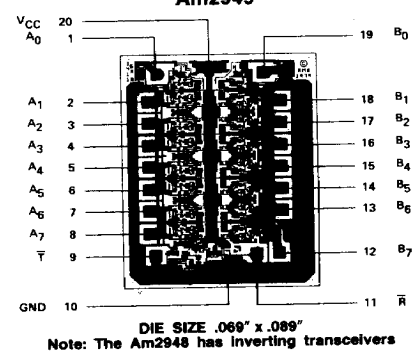


Note: Pin 1 is marked for orientation
Am2948 is inverting from Ai to Bi

LOGIC SYMBOL

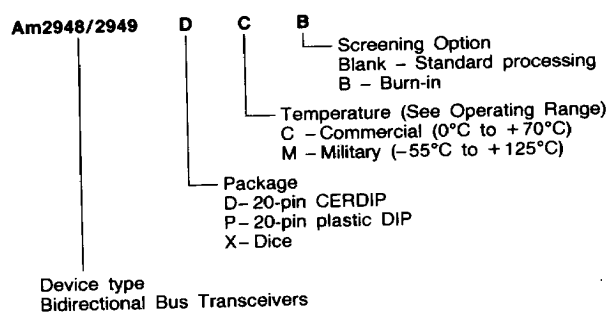


METALLIZATION AND PAD LAYOUT Am2949



ORDERING INFORMATION

AMD products are available in several packages and operating ranges. The order number is formed by a combination of the following: Device number, speed option (if applicable), package type, operating range and screening option (if desired).



Valid Combinations	
Am2948	PC
Am2949	DC, DCB, DM, DMB, XC

Valid Combinations
Consult the AMD sales office in your area to determine if a device is currently available in the combination you wish.

PIN DESCRIPTION

Pin No.	Name	I/O	Description
	A ₀ -A ₇	I/O	A port inputs/outputs are receiver output drivers when Receive is LOW and Transmit is HIGH, and are transmit inputs when Receive is HIGH and Transmit is LOW.
	B ₀ -B ₇	I/O	B port inputs/outputs are transmit output drivers when Transmit is LOW and Receive is HIGH, and are receiver inputs when Transmit is HIGH and Receive is LOW.
9,11	Transmit, Receive	I	These controls determine whether A port and B port drivers are in 3-state. With both Transmit and Receive HIGH both ports are in 3-state. Transmit and Receive both LOW activate both drivers and may cause oscillations. This is not an intended logic condition. With Transmit HIGH and Receive LOW A port is the output and B port is the input. With Transmit LOW and Receive HIGH B port is the output and A port is the input.

FUNCTION TABLE

Control Inputs		Resulting Conditions	
Transmit	Receive	A Port	B Port
H	L	Out	In
L	H	In	Out
H	H	3-State	3-State
L	L	Both Active*	

*This is not an intended logic condition and may cause oscillations.

ABSOLUTE MAXIMUM RATINGS

Storage Temperature -65°C to +150°C
 Supply Voltage 7.0V
 Input Voltage 5.5V
 Output Voltage 5.5V
 Lead Temperature (Soldering, 10 seconds) 300°C

Stresses above those listed under ABSOLUTE MAXIMUM RATINGS may cause permanent device failure. Functionality at or above these limits is not implied. Exposure to absolute maximum ratings for extended periods may affect device reliability.

OPERATING RANGES

Commercial (C) Devices
 Temperature 0°C to +70°C
 Supply Voltage +4.75V to +5.25V

Military (M) Devices
 Temperature -55°C to +125°C
 Supply Voltage +4.5V to +5.5V

Operating ranges define those limits over which the functionality of the device is guaranteed.

DC CHARACTERISTICS over operating range unless otherwise specified

Parameters	Description	Test Conditions		Min	Typ	Max	Units
A PORT (A₀-A₇)							
V _{IH}	Logical "1" Input Voltage	T̄ = 0.8V, R̄ = 2.0V		2.0			Volts
V _{IL}	Logical "0" Input Voltage	T̄ = 0.8V, R̄ = 2.0V		COM'L MIL		0.8 0.7	Volts
V _{OH}	Logical "1" Output Voltage	T̄ = 2.0V, R̄ = 0.8V	I _{OH} = -0.4mA I _{OH} = -3.0mA	V _{CC} - 1.15 2.7	V _{CC} - 0.7 3.95		Volts
V _{OL}	Logical "0" Output Voltage	T̄ = 2.0V, R̄ = 0.8V	I _{OL} = 12mA COM'L I _{OL} = 24mA	0.3 0.35	0.4 0.50		Volts
I _{OS}	Output Short Circuit Current	T̄ = 2.0V, R̄ = 0.8V, V _O = 0V, V _{CC} = MAX; Note 2		-10	-38	-75	mA
I _{IH}	Logical "1" Input Current	T̄ = 0.8V, R̄ = 2.0V, V _I = 2.7V			0.1	80	μA
I _I	Input Current at Maximum Input Voltage	T̄ = R̄ = 2.0V, V _{CC} = MAX, V _I = V _{CC} MAX				1	mA
I _{IL}	Logical "0" Input Current	T̄ = 0.8V, R̄ = 2.0V, V _I = 0.4V			-70	-200	μA
V _C	Input Clamp Voltage	T̄ = R̄ = 2.0V, I _{IN} = -12mA			-0.7	-1.5	Volts
I _{OD}	Output/Input 3-State Current	T̄ = R̄ = 2.0V		V _O = 0.4V V _O = 4.0V		-200 80	μA
B PORT (B₀-B₇)							
V _{IH}	Logical "1" Input Voltage	T̄ = 2.0V, R̄ = 0.8V		2.0			Volts
V _{IL}	Logical "0" Input Voltage	T̄ = 2.0V, R̄ = 0.8V		COM'L MIL		0.8 0.7	Volts
V _{OH}	Logical "1" Output Voltage	T̄ = 0.8V, R̄ = 2.0V	I _{OH} = -0.4mA I _{OH} = -5.0mA I _{OH} = -10mA	V _{CC} - 1.15 2.7 2.4	V _{CC} - 0.8 3.9 3.6		Volts
V _{OL}	Logical "0" Output Voltage	T̄ = 0.8V, R̄ = 2.0V	I _{OL} = 20mA I _{OL} = 48mA		0.3 0.4	0.4 0.5	Volts
I _{OS}	Output Short Circuit Current	T̄ = 0.8V, R̄ = 2.0V, V _O = 0V V _{CC} = MAX; Note 2		-25	-50	-150	mA
I _{IH}	Logical "1" Input Current	T̄ = 2.0V, R̄ = 0.8V, V _I = 2.7V			0.1	80	μA
I _I	Input Current at Maximum Input Voltage	T̄ = R̄ = 2.0V, V _{CC} = MAX, V _I = V _{CC} MAX				1	mA
I _{IL}	Logical "0" Input Current	T̄ = 2.0V, R̄ = 0.8V, V _I = 0.4V			-70	-200	μA
V _C	Input Clamp Voltage	T̄ = R̄ = 2.0V, I _{IN} = -12mA			-0.7	-1.5	Volts
I _{OD}	Output/Input 3-State Current	T̄ = R̄ = 2.0V		V _O = 0.4V V _O = 4.0V		-200 200	μA

5

DC CHARACTERISTICS (Cont.)

CONTROL INPUTS $\bar{T}$, $\bar{R}$

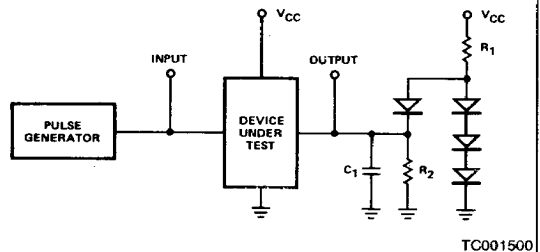
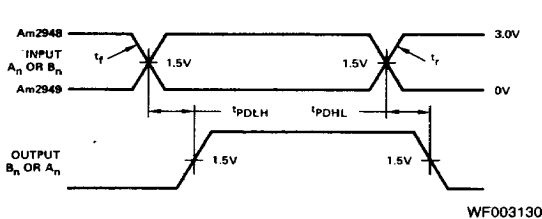
V_{IH}	Logical "1" Input Voltage			2.0			Volts
V_{IL}	Logical "0" Input Voltage		COM'L			0.8	Volts
			MIL			0.7	Volts
I_{IH}	Logical "1" Input Current	$V_I = 2.7V$			0.5	20	μA
I_I	Input Current at Maximum Input Voltage	$V_{CC} = \text{MAX}, V_I = V_{CC} \text{ MAX}$				1.0	mA
I_{IL}	Logical "0" Input Current	$V_I = 0.4V$	$\bar{R}$		-0.1	-0.25	mA
			$\bar{T}$		-0.25	-0.5	mA
V_C	Input Clamp Voltage	$I_{IN} = -12mA$			-0.8	-1.5	Volts

POWER SUPPLY CURRENT

I_{CC}	Power Supply Current	Am2948	$\bar{T} = \bar{R} = 2.0V, V_I = 2.0V, V_{CC} = \text{MAX}$		70	100	mA
			$\bar{T} = 0.4V, V_{INA} = \bar{R} = 2.0V, V_{CC} = \text{MAX}$		100	150	
		Am2949	$\bar{T} = \bar{R} = 2.0V, V_I = 0.4V, V_{CC} = \text{MAX}$		70	100	mA
			$\bar{T} = V_{INA} = 0.4V, \bar{R} = 2.0V, V_{CC} = \text{MAX}$		90	140	

SWITCHING TEST WAVEFORM

SWITCHING TEST CIRCUIT

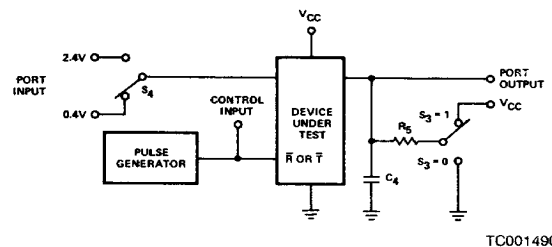
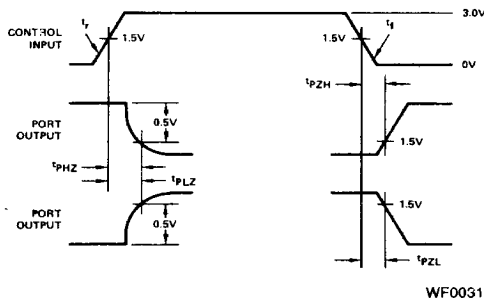


$$t_r = t_f < 10\text{ns}$$

$$10\% \text{ to } 90\%$$

Note: C_1 includes test fixture capacitance.

Figure A. Propagation Delay from A Port to B Port or from B Port to A Port.



$$t_r = t_f < 10\text{ns}$$

$$10\% \text{ to } 90\%$$

Note: C_4 includes test fixture capacitance. Port input is in a fixed logical condition. See AC table.Figure B. Propagation Delay to/from Three-State from $\bar{R}$ to A Port and $\bar{T}$ to B Port.

SWITCHING CHARACTERISTICS ($T_A = +25^\circ\text{C}$, $V_{CC} = 5.0\text{V}$)
Am2948

Parameter	Description	Test Conditions	Typ	Max	Units
A PORT DATA/MODE SPECIFICATIONS					
t_{PDHLA}	Propagation Delay to a Logical "0" from B Port to A Port	$\bar{T} = 2.4\text{V}$, $\bar{R} = 0.4\text{V}$ (Figure A) $R_1 = 1\text{k}$, $R_2 = 5\text{k}$, $C_1 = 30\text{pF}$	8	12	ns
t_{PDLHA}	Propagation Delay to a Logical "1" from B Port to A Port	$\bar{T} = 2.4\text{V}$, $\bar{R} = 0.4\text{V}$ (Figure A) $R_1 = 1\text{k}$, $R_2 = 5\text{k}$, $C_1 = 30\text{pF}$	11	16	ns
t_{PLZA}	Propagation Delay from a Logical "0" to 3-State from $\bar{R}$ to A Port	B_0 to $B_7 = 2.4\text{V}$, $\bar{T} = 2.4\text{V}$ (Figure B) $S_3 = 1$, $R_5 = 1\text{k}$, $C_4 = 15\text{pF}$	10	15	ns
t_{PHZA}	Propagation Delay from a Logical "1" to 3-State from $\bar{R}$ to A Port	B_0 to $B_7 = 0.4\text{V}$, $\bar{T} = 2.4\text{V}$ (Figure B) $S_3 = 0$, $R_5 = 1\text{k}$, $C_4 = 15\text{pF}$	8	15	ns
t_{PZLA}	Propagation Delay from 3-State to a Logical "0" from $\bar{R}$ to A Port	B_0 to $B_7 = 2.4\text{V}$, $\bar{T} = 2.4\text{V}$ (Figure B) $S_3 = 1$, $R_5 = 1\text{k}$, $C_4 = 30\text{pF}$	20	27	ns
t_{PZHA}	Propagation Delay from 3-State to a Logical "1" from $\bar{R}$ to A Port	B_0 to $B_7 = 0.4\text{V}$, $\bar{T} = 2.4\text{V}$ (Figure B) $S_3 = 0$, $R_5 = 1\text{k}$, $C_4 = 30\text{pF}$	20	27	ns
B PORT DATA/MODE SPECIFICATIONS					
t_{PDHLB}	Propagation Delay to a Logical "0" from A Port to B Port	$\bar{T} = 0.4\text{V}$, $\bar{R} = 2.4\text{V}$ (Figure A) $R_1 = 100\Omega$, $R_2 = 1\text{k}$, $C_1 = 300\text{pF}$	12	18	ns
		$R_1 = 667\Omega$, $R_2 = 5\text{k}$, $C_1 = 45\text{pF}$	8	12	ns
t_{PDLHB}	Propagation Delay to a Logical "1" from A Port to B Port	$\bar{T} = 0.4\text{V}$, $\bar{R} = 2.4\text{V}$ (Figure A) $R_1 = 100\Omega$, $R_2 = 1\text{k}$, $C_1 = 300\text{pF}$	15	20	ns
		$R_1 = 667\Omega$, $R_2 = 5\text{k}$, $C_1 = 45\text{pF}$	9	14	ns
t_{PLZB}	Propagation Delay from a Logical "0" to 3-State from $\bar{T}$ to B Port	A_0 to $A_7 = 2.4\text{V}$, $\bar{R} = 2.4\text{V}$ (Figure B) $S_3 = 1$, $R_5 = 1\text{k}$, $C_4 = 15\text{pF}$	13	18	ns
t_{PHZB}	Propagation Delay from a Logical "1" to 3-State from $\bar{T}$ to B Port	A_0 to $A_7 = 0.4\text{V}$, $\bar{R} = 2.4\text{V}$ (Figure B) $S_3 = 0$, $R_5 = 1\text{k}$, $C_4 = 15\text{pF}$	8	15	ns
t_{PZLB}	Propagation Delay from 3-State to a Logical "0" from $\bar{T}$ to B Port	A_0 to $A_7 = 2.4\text{V}$, $\bar{R} = 2.4\text{V}$ (Figure B) $S_3 = 1$, $R_5 = 100\Omega$, $C_4 = 300\text{pF}$	25	35	ns
		$S_3 = 1$, $R_5 = 667\Omega$, $C_1 = 45\text{pF}$	18	25	ns
t_{PZHB}	Propagation Delay from 3-State to a Logical "1" from $\bar{T}$ to B Port	A_0 to $A_7 = 0.4\text{V}$, $\bar{R} = 2.4\text{V}$ (Figure B) $S_3 = 0$, $R_5 = 1\text{k}$, $C_4 = 300\text{pF}$	25	35	ns
		$S_3 = 0$, $R_5 = 5\text{k}$, $C_4 = 45\text{pF}$	16	25	ns

5

SWITCHING CHARACTERISTICS over operating range unless otherwise specified
Am2948

Parameter	Description	Test Conditions	COMMERCIAL Am2948	MILITARY Am2948	Units
			Max	Max	
A PORT DATA/MODE SPECIFICATIONS					
t _{PDHLA}	Propagation Delay to a Logical "0" from B Port to A Port	T = 2.4V, $\bar{R}$ = 0.4V (Figure A) R ₁ = 1k, R ₂ = 5k, C ₁ = 30pF	19	16	ns
t _{PDLHA}	Propagation Delay to a Logical "1" from B Port to A Port	T = 2.4V, $\bar{R}$ = 0.4V (Figure A) R ₁ = 1k, R ₂ = 5k, C ₁ = 30pF	23	20	ns
t _{PLZA}	Propagation Delay from a Logical "0" to 3-State from R to A Port	B ₀ to B ₇ = 2.4V, T = 2.4V (Figure B) S ₃ = 1, R ₅ = 1k, C ₄ = 15pF	21	18	ns
t _{PHZA}	Propagation Delay from a Logical "1" to 3-State from R to A Port	B ₀ to B ₇ = 0.4V, T = 2.4V (Figure B) S ₃ = 0, R ₅ = 1k, C ₄ = 15pF	21	18	ns
t _{PZLA}	Propagation Delay from 3-State to a Logical "0" from R to A Port	B ₀ to B ₇ = 2.4V, T = 2.4V (Figure B) S ₃ = 1, R ₅ = 1k, C ₄ = 30pF	35	30	ns
t _{PZHA}	Propagation Delay from 3-State to a Logical "1" from R to A Port	B ₀ to B ₇ = 0.4V, T = 2.4V (Figure B) S ₃ = 0, R ₅ = 5k, C ₄ = 30pF	35	30	ns
B PORT DATA/MODE SPECIFICATIONS					
t _{PDHLB}	Propagation Delay to a Logical "0" from A Port to B Port	T = 0.4V, $\bar{R}$ = 2.4V (Figure A)	29	24	ns
		R ₁ = 100Ω, R ₂ = 1k, C ₁ = 300pF R ₁ = 667Ω, R ₂ = 5k, C ₁ = 45pF	19	16	ns
t _{PDLHB}	Propagation Delay to a Logical "1" from A Port to B Port	T = 0.4V, $\bar{R}$ = 2.4V (Figure A)	30	25	ns
		R ₁ = 100Ω, R ₂ = 1k, C ₁ = 300pF R ₁ = 667Ω, R ₂ = 5k, C ₁ = 45pF	22	19	ns
t _{PLZB}	Propagation Delay from a Logical "0" to 3-State from T to B Port	A ₀ to A ₇ = 2.4V, $\bar{R}$ = 2.4V (Figure B) S ₃ = 1, R ₅ = 1k, C ₄ = 15pF	26	23	ns
t _{PHZB}	Propagation Delay from a Logical "1" to 3-State from T to B Port	A ₀ to A ₇ = 0.4V, $\bar{R}$ = 2.4V (Figure B) S ₃ = 0, R ₅ = 1k, C ₄ = 15pF	21	18	ns
t _{PZLB}	Propagation Delay from 3-State to a Logical "0" from T to B Port	A ₀ to A ₇ = 2.4V, $\bar{R}$ = 2.4V (Figure B) S ₃ = 1, R ₅ = 100Ω, C ₄ = 300pF	43	38	ns
		S ₃ = 1, R ₅ = 667Ω, C ₄ = 45pF	33	28	ns
t _{PZHB}	Propagation Delay from 3-State to a Logical "1" from T to B Port	A ₀ to A ₇ = 0.4V, $\bar{R}$ = 2.4V (Figure B) S ₃ = 0, R ₅ = 1k, C ₄ = 300pF	43	38	ns
		S ₃ = 0, R ₅ = 5k, C ₄ = 45pF	33	28	ns

SWITCHING CHARACTERISTICS ($T_A = +25^\circ\text{C}$, $V_{CC} = 5.0\text{V}$)
Am2949

Parameter	Description	Test Conditions	Typ	Max	Units
A PORT DATA/MODE SPECIFICATIONS					
t_{PDHLA}	Propagation Delay to a Logical "0" from B Port to A Port	$\bar{T} = 2.4\text{V}$, $\bar{R} = 0.4\text{V}$ (Figure A) $R_1 = 1\text{k}$, $R_2 = 5\text{k}$, $C_1 = 30\text{pF}$	14	18	ns
t_{PDLHA}	Propagation Delay to a Logical "1" from B Port to A Port	$\bar{T} = 2.4\text{V}$, $\bar{R} = 0.4\text{V}$ (Figure A) $R_1 = 1\text{k}$, $R_2 = 5\text{k}$, $C_1 = 30\text{pF}$	13	18	ns
t_{PLZA}	Propagation Delay from a Logical "0" to 3-State from $\bar{R}$ to A Port	B_0 to $B_7 = 0.4\text{V}$, $\bar{T} = 2.4\text{V}$ (Figure B) $S_3 = 1$, $R_5 = 1\text{k}$, $C_4 = 15\text{pF}$	11	15	ns
t_{PHZA}	Propagation Delay from a Logical "1" to 3-State from $\bar{R}$ to A Port	B_0 to $B_7 = 2.4\text{V}$, $\bar{T} = 2.4\text{V}$ (Figure B) $S_3 = 0$, $R_5 = 1\text{k}$, $C_4 = 15\text{pF}$	8	15	ns
t_{PZLA}	Propagation Delay from 3-State to a Logical "0" from $\bar{R}$ to A Port	B_0 to $B_7 = 0.4\text{V}$, $\bar{T} = 2.4\text{V}$ (Figure B) $S_3 = 1$, $R_5 = 1\text{k}$, $C_4 = 30\text{pF}$	20	27	ns
t_{PZHA}	Propagation Delay from 3-State to a Logical "1" from $\bar{R}$ to A Port	B_0 to $B_7 = 2.4\text{V}$, $\bar{T} = 2.4\text{V}$ (Figure B) $S_3 = 1$, $R_5 = 5\text{k}$, $C_4 = 30\text{pF}$	20	27	ns
B PORT DATA/MODE SPECIFICATIONS					
t_{PDHLB}	Propagation Delay to a Logical "0" from A Port to B Port	$\bar{T} = 0.4\text{V}$, $\bar{R} = 2.4\text{V}$ (Figure A) $R_1 = 100\Omega$, $R_2 = 1\text{k}$, $C_1 = 300\text{pF}$	18	23	ns
		$R_1 = 667\Omega$, $R_2 = 5\text{k}$, $C_1 = 45\text{pF}$	11	18	ns
t_{PDLHB}	Propagation Delay to a Logical "1" from A Port to B Port	$\bar{T} = 0.4\text{V}$, $\bar{R} = 2.4\text{V}$ (Figure A) $R_1 = 100\Omega$, $R_2 = 1\text{k}$, $C_1 = 300\text{pF}$	16	23	ns
		$R_1 = 667\Omega$, $R_2 = 5\text{k}$, $C_1 = 45\text{pF}$	11	18	ns
t_{PLZB}	Propagation Delay from a Logical "0" to 3-State from $\bar{T}$ to B Port	A_0 to $A_7 = 0.4\text{V}$, $\bar{R} = 2.4\text{V}$ (Figure B) $S_3 = 1$, $R_5 = 1\text{k}$, $C_4 = 15\text{pF}$	13	18	ns
t_{PHZB}	Propagation Delay from a Logical "1" to 3-State from $\bar{T}$ to B Port	A_0 to $A_7 = 2.4\text{V}$, $\bar{R} = 2.4\text{V}$ (Figure B) $S_3 = 0$, $R_5 = 1\text{k}$, $C_4 = 15\text{pF}$	8	15	ns
t_{PZLB}	Propagation Delay from 3-State to a Logical "0" from $\bar{T}$ to B Port	A_0 to $A_7 = 0.4\text{V}$, $\bar{R} = 2.4\text{V}$ (Figure B) $S_3 = 1$, $R_5 = 100\Omega$, $C_4 = 300\text{pF}$	25	35	ns
		$S_3 = 1$, $R_5 = 667\Omega$, $C_1 = 45\text{pF}$	17	25	ns
t_{PZHB}	Propagation Delay from 3-State to a Logical "1" from $\bar{T}$ to B Port	A_0 to $A_7 = 2.4\text{V}$, $\bar{R} = 2.4\text{V}$ (Figure B) $S_3 = 0$, $R_5 = 1\text{k}$, $C_4 = 300\text{pF}$	24	35	ns
		$S_3 = 0$, $R_5 = 5\text{k}$, $C_1 = 45\text{pF}$	17	25	ns

5

SWITCHING CHARACTERISTICS over operating range unless otherwise specified
Am2949

Parameter	Description	Test Conditions	COMMERCIAL Am2949	MILITARY Am2949	Units
			Max	Max	
A PORT DATA/MODE SPECIFICATIONS					
t _{PDHLA}	Propagation Delay to a Logical "0" from B Port to A Port	T = 2.4V, R = 0.4V (Figure A) R ₁ = 1k, R ₂ = 5k, C ₁ = 30pF	24	21	ns
t _{PDLHA}	Propagation Delay to a Logical "1" from B Port to A Port	T = 2.4V, R = 0.4V (Figure A) R ₁ = 1k, R ₂ = 5k, C ₁ = 30pF	24	21	ns
t _{PLZA}	Propagation Delay from a Logical "0" to 3-State from R to A Port	B ₀ to B ₇ = 0.4V, T = 2.4V (Figure B) S ₃ = 1, R ₅ = 1k, C ₄ = 15pF	21	18	ns
t _{PHZA}	Propagation Delay from a Logical "1" to 3-State from R to A Port	B ₀ to B ₇ = 2.4V, T = 2.4V (Figure B) S ₃ = 0, R ₅ = 1k, C ₄ = 15pF	21	18	ns
t _{PZLA}	Propagation Delay from 3-State to a Logical "0" from R to A Port	B ₀ to B ₇ = 0.4V, T = 2.4V (Figure B) S ₃ = 1, R ₅ = 1k, C ₄ = 30pF	35	30	ns
t _{PZHA}	Propagation Delay from 3-State to a Logical "1" from R to A Port	B ₀ to B ₇ = 2.4V, T = 2.4V (Figure B) S ₃ = 0, R ₅ = 5k, C ₄ = 30pF	35	30	ns
B PORT DATA/MODE SPECIFICATIONS					
t _{PDHLB}	Propagation Delay to a Logical "0" from A Port to B Port	T = 0.4V, R = 2.4V (Figure A)	34	28	ns
		R ₁ = 100Ω, R ₂ = 1k, C ₁ = 300pF R ₁ = 667Ω, R ₂ = 5k, C ₁ = 45pF	25	22	ns
t _{PDLHB}	Propagation Delay to a Logical "1" from A Port to B Port	T = 0.4V, R = 2.4V (Figure A)	34	28	ns
		R ₁ = 100Ω, R ₂ = 1k, C ₁ = 300pF R ₁ = 667Ω, R ₂ = 5k, C ₁ = 45pF	25	22	ns
t _{PLZB}	Propagation Delay from a Logical "0" to 3-State from T to B Port	A ₀ to A ₇ = 0.4V, R = 2.4V (Figure B) S ₃ = 1, R ₅ = 1k, C ₄ = 15pF	26	23	ns
t _{PHZB}	Propagation Delay from a Logical "1" to 3-State from T to B Port	A ₀ to A ₇ = 2.4V, R = 2.4V (Figure B) S ₃ = 0, R ₅ = 1k, C ₄ = 15pF	21	18	ns
t _{PZLB}	Propagation Delay from 3-State to a Logical "0" from T to B Port	A ₀ to A ₇ = 0.4V, R = 0.4V (Figure B) S ₃ = 1, R ₅ = 100Ω, C ₄ = 300pF	43	38	ns
		S ₃ = 1, R ₅ = 667Ω, C ₄ = 45pF	33	28	ns
t _{PZHB}	Propagation Delay from 3-State to a Logical "1" from T to B Port	A ₀ to A ₇ = 2.4V, R = 2.4V (Figure B) S ₃ = 0, R ₅ = 1k, C ₄ = 300pF	43	38	ns
		S ₃ = 0, R ₅ = 5k, C ₄ = 45pF	33	28	ns